

TABLE II
Minimum Distance between Different Keys as a Matrix

Switch type	Push button	Push button array	Legend switch	Slide and rocker switch	Toggle switch	Thumb wheel	Rotary switch	Large J handle	Small J handle
Push button	13	50	50	13	13	13	13	152	76
Push button array	50	50	50	38	38	38	50	152	76
Legend switch	50	50	50	38	38	38	50	152	76
Slide and rocker switch	13	38	38	13	19	13	13	127	50
Toggle switch	13	38	38	19	19	13	19	152	76
Thumb wheel	13	38	38	13	13	13	19	127	50
Rotary switch	13	50	50	13	19	19	25	127	50
Large J handle	152	152	152	127	152	127	127	76	127
Small J handle	76	76	76	50	76	50	50	127	25

Note: All measurements are in mm

should be selected (if the environment is dusty or humid, a sealed switch should be used). These come in different sizes and with/without tactile feedback. Since these keys typically come in two parts, main switch and key top, customisation and changes can be easily implemented.

Membrane keys. Membrane keys are the sleekest and can only be custom-designed. While their design is fairly straightforward in terms of usage, these have some restrictions. Since these keys are very thin when placed in a keyboard, it is hard to identify different keys (so old people or visually-challenged users will find it difficult to use instruments having such keys). Also, tactile feedback has to be incorporated in the design itself, and sometimes this needs special processes, as plastic films have to be moulded.

One big challenge is the amount of force that is needed to operate these keys, which is much higher than mechanical keys. These keys are prone to fail when tactile feedback is absent.

Touch keys. These keys are formed with a combination of hardware and software. Typically, the display itself becomes a keyboard with a touch-key overlay and graphics displayed in the display.

There are three types of touch technologies that are in use: 4-wire resistive, 5-wire resistive and capacitive. Table III captures their charac-

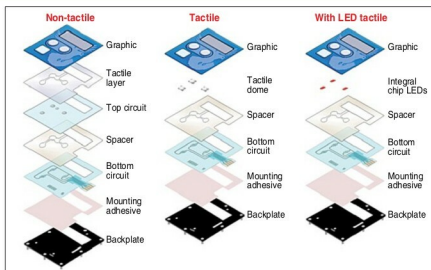


Fig. 4: Types of membrane keys

teristics for a better understanding.

Some advantages of the touchscreen are:

1. Points directly to objects; there is a direct relationship between hand and cursor movements (distance, speed and direction)
2. Intuitive to use
3. Since the hand is moving on the same surface as the cursor, manipulating objects on the screen is similar to manipulating these in the manual world
4. Fast, but less precise without a pen
5. A finger can be used as well, apart from a pen (usually no cable is needed)
6. No keyboard is necessary for applications that need menu selec-

tions only, and this also saves desk space

7. Suitable for novices, applications for information retrieval and high-use environments

8. Is adaptable; since the keyboard layout is software-generated, it is much easier to support different languages (one reason why ATMs use touchscreens)

There are some disadvantages too, which are as follows:

1. Low-precision (finger): Imprecise positioning, possible problems with eye-parallel axis (with pen, too); the finger may be too large for accurately pointing at small objects (a pen is more accurate)
2. Hand movements (if used with a keyboard): Requires the user to